Graphics Programming

Spheres and Ellipsoids

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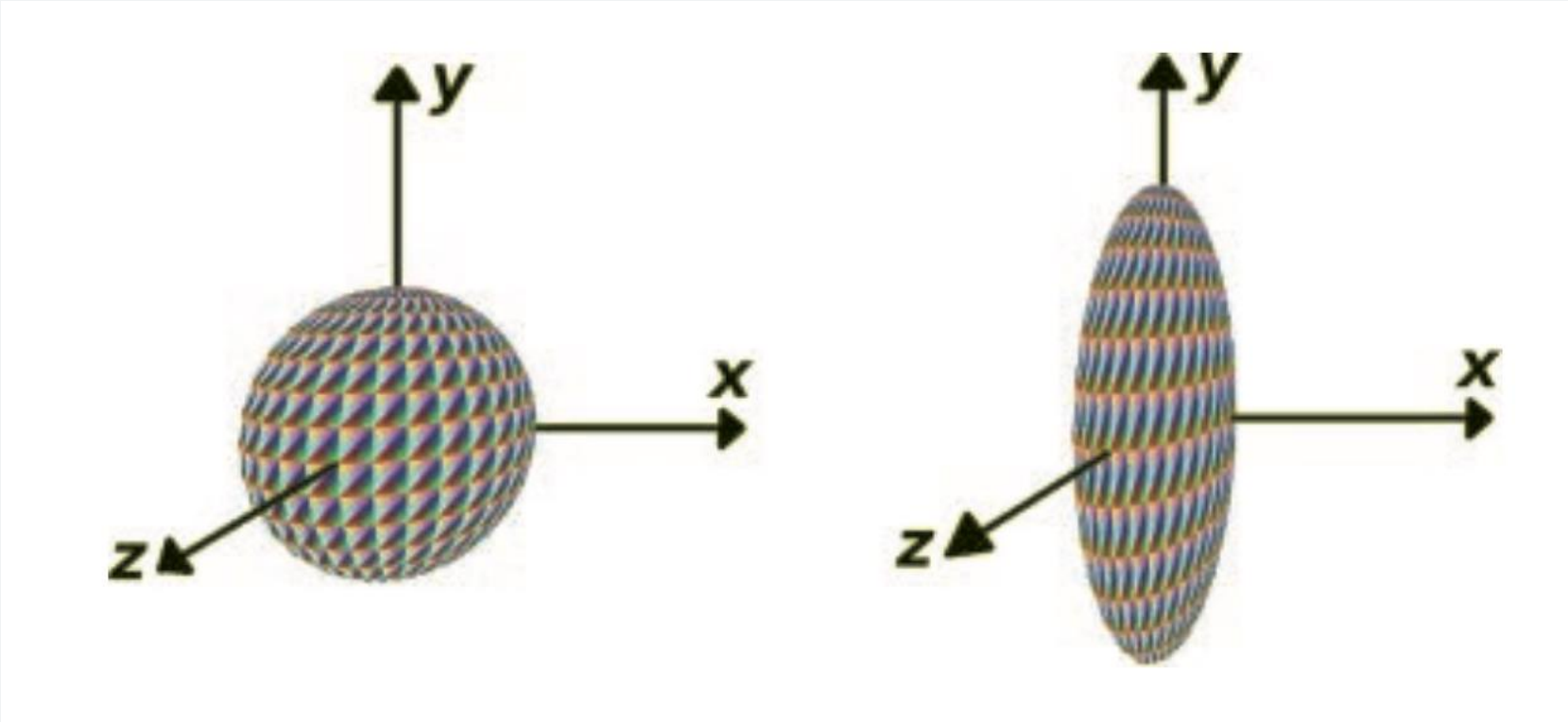

Parametric Equation of a Sphere (1)

- Assume the sphere is centered at the origin and has radius 1
- Think of sphere as multiple cross-sections alone y-axis
  - Cross-sections of a sphere are circles
  - Then, the problem of drawing a sphere became drawing multiple circles.
    - Parametric equation of a circle
      - $x = R \cdot cos(t)$ and $y = R \cdot sin(t)$

Parametric Equation of a Sphere (2)

- Figure out X and Z value of a cross-section
  - 2D circle is on x, y plan, however cross-sections of a sphere are on the x, z plan. We need to figure out the x, z coordinates for a sphere
  - $z = R \cdot cos(u)$ and $x = R \cdot sin(u)$, where $0 \leq u \leq 2pi$
    - R is the radius of the cross-section, not the radius of the sphere.
    - R is NOT a constant value! R value depends on where the circle is in the sphere.
    - R is related to Y value

Parametric Equation of a Sphere (3)

- Figure out Y value of a cross-section
  - Since $x^2+y^2+z^2=1$, substitute x and z equations: $R^2+y^2=1$
- Let $R=cos(v)$ and $y=sin(v)$
  - Since Y value must be between -1 and 1, then $-pi/2 \leq v \leq pi/2$

- Result: (x,y,z) = (sin(u)•cos(v), sin(v), cos(u)•cos(v))